

AnaToPrint: Agile Software Development for 3D Printing of Medical Data

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Our capstone project presents a client-side web application that converts medical CT scan data (DICOM format) into files suitable for 3D printing, addressing the need for accessible and privacy-preserving medical imaging tools. Existing solutions often rely on server-side processing, raising concerns about data security and limiting accessibility. In contrast, this application performs all processing locally within the user's web browser, ensuring that sensitive medical data remains private while maintaining cross-platform usability.

Our group spent time working on this through the agile development process. We made sure to constantly iterate as we went and have working code at all times that we could test and run as we developed. We wanted to keep ourselves ready to go at all times and make sure that any changes needed could be made at a moment's notice. Our group held weekly scrum meetings to make sure that we as a group were all on the same page and knew what each of us had to do for the week. Development like this was a new practice for some; however, we all were able to work seamlessly together and create a project I believe we are all proud of.

The application successfully generates printable .stl models and functional .gcode outputs, demonstrating the feasibility of fully client-side medical image processing. These results directly support the project's goal of enabling secure, accessible 3D model generation from CT data. This work contributes to the growing intersection of web technologies and biomedical engineering, with potential applications in education, research, and low-cost medical prototyping.

The system accepts user-submitted DICOM files and reconstructs them into a 3D voxel volume with associated spatial metadata. Users select a target tissue type, either bone, skin, or muscle, and the application will apply Hounsfield Unit-based thresholding (HU) to segment the desired anatomy. An interactive 3D preview enables real-time visualization and adjustment, supporting accurate and user-guided refinement. Surface extraction is performed using a Marching Cubes algorithm to generate polygonal meshes, while a custom pipeline produces .gcode that encodes both geometry and density-mimicking properties. This methodology was selected to balance computational efficiency with anatomical fidelity in a browser-based environment.